Spectral Analysis of the Neumann Laplacian on a Dumbbell Domain
Theory and Finite Element Simulations

Abou Bakr Al Ajami
M2 MMS – Université de Pau et des Pays de l'Adour
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Contents

1 Introduction

In recent years, there has been an increasing interest in understanding how geometry affects wave propagation through spectral analysis of partial differential equations. In this project, we study the Laplace operator with homogeneous Neumann boundary conditions on a bounded domain $\Omega \subset \mathbb{R}^2$.

The objective of this project is as follows: First, we analyze theoretically the spectral properties of the Neumann Laplacian and show how its eigenvalues and eigenfunctions provide a natural decomposition of the forced wave equation. Second, we approximate these quantities numerically using the finite element method [3] and study the dynamical response of the system.

The domain has a dumbbell shape, consisting of two cavities connected by a corridor. This geometry is particularly relevant as it may produce mode localization.

By testing the system with frequencies of the form $\omega = \sqrt{\lambda_n}$, we investigate resonance phenomena and identify the dominant modes.

2 Functional Analysis

Let $\Omega \subset \mathbb{R}^2$ be a bounded, connected, Lipschitz domain.

We define

$$H = L^2(\Omega), \qquad V = H^1(\Omega).$$

We consider the Neumann eigenvalue problem

$$\begin{cases} -\Delta\phi = \lambda\phi & \text{in } \Omega, \\ \partial_n\phi = 0 & \text{on } \partial\Omega. \end{cases}$$

We introduce the bilinear forms

$$a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx, \qquad m(u, v) = \int_\Omega uv \, dx, \qquad u, v \in H^1(\Omega).$$

The form $a(\cdot, \cdot)$ is symmetric and continuous on $H^1(\Omega)$ Thanks to cauchy shcawrsz's inequalit, while $m(\cdot, \cdot)$ is the standard inner product of $L^2(\Omega)$.

2.1 Weak formulation

Assume first that ϕ is smooth. Multiplying the equation $-\Delta\phi = \lambda\phi$ by a test function $v \in H^1(\Omega)$ and integrating over Ω, we obtain

$$\int_\Omega (-\Delta\phi) \, v \, dx = \lambda \int_\Omega \phi v \, dx.$$

Using Green's formula,

$$\int_\Omega \nabla\phi \cdot \nabla v \, dx - \int_{\partial\Omega} \partial_n\phi \, v \, ds = \lambda \int_\Omega \phi v \, dx.$$

Since $\partial_n\phi = 0$ on $\partial\Omega$, the boundary term vanishes, and the weak formulation becomes

$$a(\phi, v) = \lambda \, m(\phi, v) \qquad \forall v \in H^1(\Omega).$$

Thus the weak eigenvalue problem is:
Find $(\phi, \lambda) \in H^1(\Omega) \times \mathbb{R}$ with $\phi \neq 0$ such that

$$a(\phi, v) = \lambda \, m(\phi, v) \qquad \forall v \in H^1(\Omega).$$

2.2 Why introducing the mean-zero space?

We now explain why the space $H^1_*(\Omega)$ naturally appears in the Neumann problem.

Let u be a solution of the Neumann problem

$$-\Delta u = f \quad \text{in } \Omega, \qquad \partial_n u = 0 \quad \text{on } \partial\Omega.$$

Multiplying the equation by a test function $v \in H^1(\Omega)$ and integrating over Ω gives

$$\int_\Omega (-\Delta u) v \, dx = \int_\Omega f v \, dx.$$

Using Green's formula,

$$\int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} \partial_n u \, v \, ds = \int_\Omega f v \, dx.$$

Since $\partial_n u = 0$ on $\partial\Omega$, we obtain the weak formulation

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall v \in H^1(\Omega).$$

Choosing the test function $v = 1$ (note that $1 \in H^1(\Omega)$ and $\nabla 1 = 0$) yields

$$\int_\Omega \nabla u \cdot \nabla 1 \, dx = \int_\Omega f \, dx.$$

Since $\nabla 1 = 0$, it follows that

$$0 = \int_\Omega f \, dx.$$

This shows that the Neumann problem admits a solution only if the compatibility condition

$$\int_\Omega f \, dx = 0$$

is satisfied.

We therefore introduce the space

$$L^2_*(\Omega) = \left\{ f \in L^2(\Omega) : \int_\Omega f \, dx = 0 \right\}.$$

Moreover, the solution is not unique. Indeed, if u is a solution and $c \in \mathbb{R}$ is a constant, define

$$\tilde{u} = u + c.$$

Since $\nabla c = 0$, we have

$$\nabla \tilde{u} = \nabla u, \qquad -\Delta \tilde{u} = -\Delta u, \qquad \partial_n \tilde{u} = \partial_n u.$$

Thus $\tilde{u}$ satisfies the same Neumann problem. Therefore if u is a solution, then $u + c$ is also a solution for any constant c.

To remove this indeterminacy and ensure uniqueness, we impose

$$\int_\Omega u \, dx = 0.$$

This leads to the mean-zero space

$$H^1_*(\Omega) = \left\{ u \in H^1(\Omega) : \int_\Omega u\, dx = 0 \right\}.$$

We will work in this space in order to ensure both:

- existence of solutions,
- uniqueness of the solution.

The trivial eigenvalue

If $\phi \equiv c$ is constant, then $\nabla\phi = 0$, hence

$$a(\phi, v) = \int_\Omega \nabla\phi \cdot \nabla v\, dx = 0 \qquad \forall v \in H^1(\Omega).$$

Therefore $\lambda = 0$ is an eigenvalue, and constant functions are eigenfunctions.

Conversely, if $\phi \in H^1(\Omega)$ satisfies

$$a(\phi, v) = 0 \qquad \forall v \in H^1(\Omega),$$

then taking $v = \phi$ gives

$$a(\phi, \phi) = \int_\Omega |\nabla\phi|^2\, dx = 0.$$

Hence $\nabla\phi = 0$ almost everywhere in Ω, and since Ω is connected, ϕ is constant.

Therefore the eigenspace associated with $\lambda_0 = 0$ is exactly the space of constant functions.

2.3 The space $H^1_*(\Omega)$

The space

$$H^1_*(\Omega) = \left\{ u \in H^1(\Omega) : \int_\Omega u\, dx = 0 \right\}$$

is a closed subspace of $H^1(\Omega)$, and therefore it is itself a Hilbert space with the norm induced by $H^1(\Omega)$.

Proof

Define the linear map

$$L : H^1(\Omega) \to \mathbb{R}, \qquad L(u) = \int_\Omega u\, dx.$$

This map is continuous. Indeed, by the Cauchy–Schwarz inequality we have

$$\left| \int_\Omega u\, dx \right| \leq |\Omega|^{1/2} \|u\|_{L^2(\Omega)} \leq |\Omega|^{1/2} \|u\|_{H^1(\Omega)}.$$

Hence L is a continuous linear functional on $H^1(\Omega)$.

Observe that

$$H^1_*(\Omega) = \ker L.$$

Since the kernel of a continuous linear map between Banach spaces is closed, it follows that $H^1_*(\Omega)$ is a closed subspace of $H^1(\Omega)$.

Finally, because $H^1(\Omega)$ is a Hilbert space and every closed subspace of a Hilbert space is again a Hilbert space, we conclude that $H^1_*(\Omega)$ is itself a Hilbert space. Similarly for $L^2_*(\Omega)$ also we have the inclusion

$$H^1_*(\Omega) \subset L^2_*(\Omega),$$

since any function $u \in H^1_*(\Omega)$ belongs to $L^2(\Omega)$ and satisfies

$$\int_\Omega u\, dx = 0.$$

2.4 Poincaré–Wirtinger inequality

Let $\Omega \subset \mathbb{R}^d$ be bounded and connected. Then there exists a constant $C_P > 0$ such that

$$\|u\|_{L^2(\Omega)} \le C_P \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1_*(\Omega),$$

This follows from the classical Poincaré–Wirtinger inequality [4].

Proof. We argue by contradiction. Assume that the inequality is false. Then for every integer $n \ge 1$, there exists a function $u_n \in H^1_*(\Omega)$ such that

$$\|u_n\|_{L^2(\Omega)} > n\,\|\nabla u_n\|_{L^2(\Omega)}.$$

Define

$$v_n = \frac{u_n}{\|u_n\|_{L^2(\Omega)}}.$$

Then $v_n \in H^1_*(\Omega)$ and

$$\|v_n\|_{L^2(\Omega)} = 1, \qquad \|\nabla v_n\|_{L^2(\Omega)} < \frac{1}{n}.$$

Therefore

$$\|v_n\|^2_{H^1(\Omega)} = \|v_n\|^2_{L^2(\Omega)} + \|\nabla v_n\|^2_{L^2(\Omega)} \le 1 + 1 = 2$$

for n large enough. Hence (v_n) is bounded in $H^1(\Omega)$.

Since Ω is bounded and Lipschitz, then by the Rellich-Kondrachov theorem

$$H^1(\Omega) \hookrightarrow L^2(\Omega)$$

is compact. Thus, up to extracting a subsequence, there exists $v \in L^2(\Omega)$ such that

$$v_n \to v \qquad \text{strongly in } L^2(\Omega).$$

Moreover, since

$$\|\nabla v_n\|_{L^2(\Omega)} \to 0,$$

we obtain

$$\nabla v = 0$$

in L^2. Hence v is constant almost everywhere in Ω. Because Ω is connected, there exists a constant $c \in \mathbb{R}$ such that

$$v = c \qquad \text{a.e. in } \Omega.$$

On the other hand, each v_n belongs to $H^1_*(\Omega)$, so

$$\int_\Omega v_n\,dx = 0.$$

Passing to the limit in $L^2(\Omega)$ gives

$$\int_\Omega v\,dx = 0.$$

Since $v = c$, this implies

$$c\,|\Omega| = 0.$$

Because $|\Omega| > 0$, we conclude that $c = 0$. Therefore

$$v = 0.$$

But this is impossible, because strong convergence in $L^2(\Omega)$ also gives

$$\|v\|_{L^2(\Omega)} = \lim_{n\to\infty} \|v_n\|_{L^2(\Omega)} = 1.$$

This contradiction proves the result. □

2.5 Mean-zero spaces

We introduce the the mean-zero spaces

$$H^1_*(\Omega) = \left\{ u \in H^1(\Omega) : \int_\Omega u \, dx = 0 \right\},$$
$$L^2_*(\Omega) = \left\{ f \in L^2(\Omega) : \int_\Omega f \, dx = 0 \right\}.$$

On $H^1_*(\Omega)$, the Poincaré inequality holds:

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega)} \qquad \forall u \in H^1_*(\Omega).$$

Hence

$$\|u\|^2_{H^1(\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega)} \leq (C_P^2 + 1)\|\nabla u\|^2_{L^2(\Omega)}.$$

Therefore

$$a(u,u) = \|\nabla u\|^2_{L^2(\Omega)} \geq \frac{1}{C_P^2 + 1} \|u\|^2_{H^1(\Omega)} \qquad \forall u \in H^1_*(\Omega),$$

so a is coercive on $H^1_*(\Omega)$.

2.6 The Neumann Solution Operator

For $f \in L^2_*(\Omega)$, define the linear form

$$l_f(v) = \int_\Omega f v \, dx \qquad \forall v \in H^1_*(\Omega).$$

Proposition 1. *For every $f \in L^2_*(\Omega)$, the map $l_f : H^1_*(\Omega) \to \mathbb{R}$ is linear and continuous.*

Proof. Linearity is from the linearity of the integral: for all $\alpha, \beta \in \mathbb{R}$ and $v, w \in H^1_*(\Omega)$,

$$l_f(\alpha v + \beta w) = \int_\Omega f(\alpha v + \beta w) \, dx = \alpha \int_\Omega f v \, dx + \beta \int_\Omega f w \, dx = \alpha l_f(v) + \beta l_f(w).$$

To prove continuity, we use the Cauchy–Schwarz inequality:

$$|l_f(v)| = \left| \int_\Omega f v \, dx \right| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)}.$$

Since $v \in H^1_*(\Omega)$, the Poincaré inequality gives

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \leq C_P \|v\|_{H^1(\Omega)}.$$

Thus

$$|l_f(v)| \leq C_P \|f\|_{L^2(\Omega)} \|v\|_{H^1(\Omega)}.$$

Hence l_f is continuous on $H^1_*(\Omega)$. □

We now consider the variational problem:
Find $u \in H^1_*(\Omega)$ such that

$$a(u,v) = l_f(v) = \int_\Omega f v \, dx \qquad \forall v \in H^1_*(\Omega).$$

Proposition 2. *For every $f \in L^2_*(\Omega)$, there exists a unique $u \in H^1_*(\Omega)$ solving*

$$a(u,v) = \int_\Omega f v \, dx \qquad \forall v \in H^1_*(\Omega).$$

Proof. The bilinear form $a(\cdot,\cdot)$ is continuous on $H^1_*(\Omega)$:

$$|a(u,v)| = \left| \int_\Omega \nabla u \cdot \nabla v \, dx \right| \leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}.$$

Moreover, a is coercive on $H^1_*(\Omega)$ by the previous section, and l_f is continuous.
By Lax–Milgram theorem we get the existence and uniqueness of a solution $u \in H^1_*(\Omega)$. □

2.7 Operator T the inverse of the Neuman Laplacian

We define the Neumann solution operator

$$T : L^2_*(\Omega) \to L^2_*(\Omega), \qquad Tf = u,$$

where u is the unique mean-zero solution of

$$a(u, v) = \int_\Omega f v \, dx \qquad \forall v \in H^1_*(\Omega).$$

Proposition 3. *If Ω is connected, then the kernel of the Neumann Laplacian is exactly the space of constant functions:*

$$\ker(-\Delta_N) = \{u \in H^1(\Omega) : u \textit{ is constant}\}.$$

Proof. First, every constant function belongs to the kernel, since if $u \equiv c$, then

$$\nabla u = 0, \qquad \Delta u = 0, \qquad \partial_n u = 0.$$

Conversely, let $u \in H^1(\Omega)$ satisfy

$$-\Delta u = 0 \quad \text{in } \Omega, \qquad \partial_n u = 0 \quad \text{on } \partial\Omega.$$

Its weak formulation is

$$\int_\Omega \nabla u \cdot \nabla v \, dx = 0 \qquad \forall v \in H^1(\Omega).$$

Choosing $v = u$, we get

$$\int_\Omega |\nabla u|^2 \, dx = 0.$$

Hence

$$\nabla u = 0 \quad \text{a.e. in } \Omega.$$

Therefore u is constant almost everywhere in Ω. Since Ω is connected, this gives a single constant on the whole domain. □

Corollary 1. *On the mean-zero space $H^1_*(\Omega)$, the kernel of the Neumann Laplacian is trivial:*

$$\ker(-\Delta_N) \cap H^1_*(\Omega) = \{0\}.$$

Proof. Let $u \in \ker(-\Delta_N) \cap H^1_*(\Omega)$. By the previous proposition, u is constant, say $u \equiv c$. Since $u \in H^1_*(\Omega)$,

$$\int_\Omega u \, dx = 0.$$

Therefore

$$c|\Omega| = 0.$$

Since $|\Omega| > 0$, we obtain $c = 0$. Hence $u = 0$. □

Remark 1. *Thus, although the Neumann Laplacian is not invertible on all of $H^1(\Omega)$ because its kernel consists of constant functions, it becomes invertible after restriction to the mean-zero space. In this sense, the operator T is the inverse of the Neumann Laplacian on $L^2_*(\Omega)$.*

Proposition 4. *The operator*

$$T : L^2_*(\Omega) \to H^1_*(\Omega), \qquad Tf = u,$$

is linear and bounded.

Proof. Let $f \in L^2_*(\Omega)$ and let $u = Tf$ be the unique solution of

$$a(u,v) = \int_\Omega fv\,dx \qquad \forall v \in H^1_*(\Omega).$$

Linearity. Let $f_1, f_2 \in L^2_*(\Omega)$, and let

$$u_1 = Tf_1, \qquad u_2 = Tf_2.$$

Then for every $v \in H^1_*(\Omega)$,

$$a(u_1,v) = \int_\Omega f_1 v\,dx, \qquad a(u_2,v) = \int_\Omega f_2 v\,dx.$$

Hence, for all $\alpha, \beta \in \mathbb{R}$,

$$a(\alpha u_1 + \beta u_2, v) = \alpha a(u_1,v) + \beta a(u_2,v) = \int_\Omega (\alpha f_1 + \beta f_2) v\,dx.$$

Thus $\alpha u_1 + \beta u_2$ is the solution associated with $\alpha f_1 + \beta f_2$. By uniqueness,

$$T(\alpha f_1 + \beta f_2) = \alpha T f_1 + \beta T f_2.$$

Therefore T is linear.

Boundedness. Taking $v = u$ in the variational formulation, we get

$$a(u,u) = \int_\Omega fu\,dx.$$

That is,

$$\|\nabla u\|^2_{L^2(\Omega)} = \int_\Omega fu\,dx \le \|f\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)}.$$

Since $u \in H^1_*(\Omega)$, the Poincar inequality gives

$$\|u\|_{L^2(\Omega)} \le C_P \|\nabla u\|_{L^2(\Omega)}.$$

Therefore

$$\|\nabla u\|^2_{L^2(\Omega)} \le C_P \|f\|_{L^2(\Omega)} \|\nabla u\|_{L^2(\Omega)},$$

hence

$$\|\nabla u\|_{L^2(\Omega)} \le C \|f\|_{L^2(\Omega)}.$$

Applying again the Poincar inequality,

$$\|u\|_{L^2(\Omega)} \le C \|f\|_{L^2(\Omega)}.$$

Thus

$$\|u\|_{H^1(\Omega)} \le C \|f\|_{L^2(\Omega)}.$$

So T is bounded from $L^2_*(\Omega)$ into $H^1_*(\Omega)$.

Finally, since

$$\|u\|_{L^2(\Omega)} \le \|u\|_{H^1(\Omega)},$$

it follows that T is also bounded as an operator from $L^2_*(\Omega)$ into $L^2_*(\Omega)$. □

Proposition 5. *The operator T is self-adjoint on $L^2_*(\Omega)$.*

Proof. Let $f, g \in L^2_*(\Omega)$ and define

$$u = Tf, \qquad w = Tg.$$

Then

$$a(u,v) = \int_\Omega fv\,dx \qquad \forall v \in H^1_*(\Omega),$$

$$a(w,v) = \int_\Omega gv\,dx \qquad \forall v \in H^1_*(\Omega).$$

Choosing $v = w$ in the first identity and $v = u$ in the second, we get

$$a(u,w) = \int_\Omega fw\,dx, \qquad a(w,u) = \int_\Omega gu\,dx.$$

Since a is symmetric,

$$a(u,w) = a(w,u).$$

Hence

$$\int_\Omega f(Tg)\,dx = \int_\Omega (Tf)g\,dx.$$

Thus

$$m(f,Tg) = m(Tf,g),$$

which proves that T is self-adjoint. □

Proposition 6. *The operator T is positive.*

Proof. Let $f \in L^2_*(\Omega)$ and set $u = Tf$. Taking $v = u$ in the variational formulation gives

$$\int_\Omega f\,u\,dx = a(u,u) = \int_\Omega |\nabla u|^2\,dx \geq 0.$$

Therefore

$$m(Tf,f) = m(f,Tf) \geq 0.$$

So T is positive. □

Proposition 7. *The operator T is compact on $L^2_*(\Omega)$.*

Proof. Let $\{f_k\}$ be a bounded sequence in $L^2_*(\Omega)$ and define

$$u_k = Tf_k.$$

By the boundedness of T we have,

$$\|u_k\|_{H^1(\Omega)} \leq C\|f_k\|_{L^2(\Omega)}.$$

Hence $\{u_k\}$ is bounded in $H^1(\Omega)$.

Since Ω is bounded and Lipschitz, the embedding

$$H^1(\Omega) \hookrightarrow L^2(\Omega)$$

is compact by the Rellich–Kondrachov theorem.

Moreover, each u_k belongs to $H^1_*(\Omega)$, hence

$$\int_\Omega u_k\,dx = 0.$$

Since $u_k \to u$ strongly in $L^2(\Omega)$, by Cauchy–Schwarz:

$$\left| \int_\Omega (u_k - u)\, dx \right| \le |\Omega|^{1/2} \|u_k - u\|_{L^2(\Omega)} \to 0.$$

Therefore,

$$\int_\Omega u_k\, dx \to \int_\Omega u\, dx.$$

Since $\int_\Omega u_k\, dx = 0$ for all k, we obtain

$$\int_\Omega u\, dx = 0.$$

Hence $u \in L^2_*(\Omega)$.
Thus $\{u_k\}$ admits a convergent subsequence in $L^2_*(\Omega)$, then T is compact on $L^2_*(\Omega)$. □

2.7.1 Spectral Theorem for the Neumann Laplacian

Theorem 1. *The Neumann Laplacian admits an infinite sequence of eigenpairs*

$$(\phi_n, \lambda_n) \in H^1(\Omega) \times \mathbb{R}_+,$$

such that

$$0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \cdots, \qquad \lambda_n \to +\infty,$$

and the eigenfunctions form an orthonormal basis of $L^2(\Omega)$.

Proof. The operator T is compact, self-adjoint and positive on the Hilbert space $L^2_*(\Omega)$. By the spectral theorem for compact self-adjoint operators (see [2], Theorem 3.6.4), there exist an orthonormal basis $(\phi_n)_{n\ge1}$ of $L^2_*(\Omega)$ and positive numbers $(\mu_n)_{n\ge1}$ such that

$$T\phi_n = \mu_n \phi_n, \qquad \mu_n > 0, \qquad \mu_n \to 0.$$

By definition of T, we have for all $v \in H^1_*(\Omega)$

$$a(T\phi_n, v) = \int_\Omega \phi_n v\, dx.$$

Since $T\phi_n = \mu_n\phi_n$, this gives

$$a(\mu_n\phi_n, v) = \int_\Omega \phi_n v\, dx.$$

Hence

$$\mu_n a(\phi_n, v) = \int_\Omega \phi_n v\, dx,$$

and therefore

$$a(\phi_n, v) = \int_\Omega \frac{1}{\mu_n}\phi_n\, v\, dx \qquad \forall v \in H^1_*(\Omega).$$

Thus ϕ_n is an eigenfunction of the Neumann Laplacian with eigenvalue

$$\lambda_n = \frac{1}{\mu_n}.$$

Since $\mu_n \to 0^+$, we obtain

$$\lambda_n \to +\infty.$$

Moreover, any eigenfunction $u \in H^1(\Omega)$ associated with an eigenvalue $\lambda \neq 0$ necessarily belongs to $H^1_*(\Omega)$. Indeed, taking $v = 1$ in the weak formulation gives

$$a(u, 1) = \lambda \int_\Omega u \, dx.$$

Since $\nabla 1 = 0$, we have $a(u, 1) = 0$, hence

$$\lambda \int_\Omega u \, dx = 0.$$

Because $\lambda \neq 0$, it follows that

$$\int_\Omega u \, dx = 0.$$

Therefore $u \in H^1_*(\Omega)$.

Finally, adding the constant eigenfunction corresponding to $\lambda_0 = 0$, and using the inclusion

$$H^1_*(\Omega) \subset H^1(\Omega),$$

we obtain the full spectral family in $H^1(\Omega)$. □

Key Result

There exists a sequence of eigenvalues

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \lambda_n \to +\infty,$$

and corresponding eigenfunctions (ϕ_n) forming an orthonormal basis of $L^2(\Omega)$.

Corollary 2. *Every function $u \in L^2(\Omega)$ can be expanded as*

$$u = \sum_{n=0}^{\infty} (u, \phi_n)_{L^2(\Omega)} \, \phi_n,$$

where the series converges in $L^2(\Omega)$.

Variational characterization of the first positive eigenvalue

The first positive eigenvalue admits the Rayleigh quotient characterization (see [1], section 6.5)

$$\lambda_1 = \inf_{\substack{u \in H^1(\Omega) \setminus \{0\} \\ \int_\Omega u \, dx = 0}} \frac{\displaystyle\int_\Omega |\nabla u|^2 \, dx}{\displaystyle\int_\Omega |u|^2 \, dx}.$$

This formula shows that λ_1 measures the smallest oscillation energy among all non-constant functions. It is also particularly useful for understanding the influence of the geometry of the domain on the lowest frequencies.

Remark 2. *If Ω is not connected, then any function that is constant on each connected component of Ω satisfies $\nabla u = 0$ almost everywhere. Hence the eigenvalue $\lambda = 0$ is no longer simple: its multiplicity is equal to the number of connected components of Ω.*

More precisely, if

$$\Omega = \Omega_1 \cup \cdots \cup \Omega_m$$

is the decomposition into connected components, then the kernel of the Neumann Laplacian is

$$\ker(-\Delta_N) = \left\{ u \in H^1(\Omega) : u \text{ is constant on each } \Omega_j \right\},$$

and therefore

$$\dim \ker(-\Delta_N) = m.$$

In particular, when Ω is connected, the kernel reduces to the space of global constant functions, so $\lambda_0 = 0$ is simple.

3 Modal Decomposition of the Wave Equation

We now study the forced wave equation

$$\partial_{tt} u - \Delta u = f(x,t)$$

in Ω, with homogeneous Neumann boundary conditions

$$\partial_n u = 0 \quad \text{on } \partial\Omega,$$

Let (ϕ_n, λ_n) the eigenpairs of the Neumann Laplacian, which form an orthonormal basis of $L^2(\Omega)$.Then solution can be written as

$$u(x,t) = \sum_{n=0}^{\infty} a_n(t)\phi_n(x),$$

where $a_n(t)$ is the *modal coefficient* (modal amplitude) associated with the n-th mode.

In the same way, we expand the forcing term as

$$f(x,t) = \sum_{n=0}^{\infty} c_n(t)\phi_n(x).$$

Derivation of the coefficients $c_n(t)$

Since (ϕ_n) is an orthonormal basis of $L^2(\Omega)$, the coefficients $c_n(t)$ are obtained by projecting $f(\cdot,t)$ onto ϕ_n by the $L^2(\Omega)$ scalar product.

Multiplying the expansion by $\phi_k(x)$ and integrating over Ω, we obtain

$$\int_\Omega f(x,t)\phi_k(x)\,dx = \sum_{n=0}^{\infty} c_n(t) \int_\Omega \phi_n(x)\phi_k(x)\,dx.$$

by orthonormality of basis

$$\int_\Omega \phi_n \phi_k \,dx = \delta_{nk},$$

we obtain

$$\int_\Omega f(x,t)\phi_k(x)\,dx = c_k(t).$$

Hence,

$$c_n(t) = \int_\Omega f(x,t)\phi_n(x)\,dx.$$

Key Result

The coefficients $c_n(t)$ are the orthogonal projections of the forcing term on to the eigenmodes:

$$c_n(t) = \langle f(\cdot,t), \phi_n \rangle_{L^2(\Omega)}.$$

Remark 3. *The coefficient $c_n(t)$ describes how strongly the forcing excites the mode ϕ_n. If $c_n(t)$ is small, then the forcing has little influence on that mode, whereas large values of $c_n(t)$ correspond to strong excitation.*

In particular, since

$$c_n(t) = \int_\Omega f(x,t)\phi_n(x)\,dx,$$

it is exactly the projection of the forcing onto the eigenmode ϕ_n.

Therefore, the sequence (c_n) gives a quantitative way to determine which modes are activated by the forcing. This is especially useful in numerical simulations, where plotting $|c_n|$ or the modal amplitudes $|a_n(t)|$ helps identify the dominant modes in the response.

Derivation of the modal equations

Substituting the expansion of u into the wave equation gives

$$\partial_{tt}u(x,t) - \Delta u(x,t) = f(x,t),$$

that is,

$$\sum_{n=0}^{\infty} a_n''(t)\phi_n(x) - \sum_{n=0}^{\infty} a_n(t)\Delta\phi_n(x) = \sum_{n=0}^{\infty} c_n(t)\phi_n(x).$$

Since the eigenfunctions satisfy

$$-\Delta\phi_n = \lambda_n\phi_n,$$

we obtain

$$\sum_{n=0}^{\infty} \left(a_n''(t) + \lambda_n a_n(t)\right)\phi_n(x) = \sum_{n=0}^{\infty} c_n(t)\phi_n(x).$$

Using the orthonormality of (ϕ_n) in $L^2(\Omega)$, we identify each mode:

$$a_n''(t) + \lambda_n a_n(t) = c_n(t), \qquad \forall n \geq 0.$$

Key Result

The wave equation reduces to a countable family of independent oscillators:

$$a_n'' + \lambda_n a_n = c_n(t).$$

This decomposition is the main link between the spectral analysis of the Neumann Laplacian and the dynamical study of the wave equation. It shows that the natural frequencies of the wave problem are precisely the square roots of the Laplacian eigenvalues:

$$\omega_n = \sqrt{\lambda_n}.$$

3.1 Solution of Each Modal Equation

We now assume that the forcing has the form

$$f(x,t) = g(x)\cos(\omega t).$$

Then, by the definition of $c_n(t)$,

$$c_n(t) = \int_\Omega g(x)\phi_n(x)\,dx\ \cos(\omega t).$$

We set
$$c_n = \int_\Omega g(x)\phi_n(x)\,dx,$$
so that
$$c_n(t) = c_n \cos(\omega t).$$
Therefore, each modal coefficient satisfies
$$a_n'' + \lambda_n a_n = c_n \cos(\omega t).$$

Case 0: the zero mode $\lambda_0 = 0$

For $n = 0$, we have $\lambda_0 = 0$, hence
$$a_0''(t) = c_0 \cos(\omega t).$$
Integrating twice, we obtain
$$a_0(t) = A_0 + B_0 t - \frac{c_0}{\omega^2}\cos(\omega t).$$

Key Result

For the constant mode $\lambda_0 = 0$, the solution is
$$a_0(t) = A_0 + B_0 t - \frac{c_0}{\omega^2}\cos(\omega t).$$

Remark 4. *The mode ϕ_0 is constant in space, so it does not represent an oscillatory shape of the domain. For this reason, resonance is not associated with the mode $\lambda_0 = 0$.*

Homogeneous solution for $\lambda_n > 0$

We now consider $n \geq 1$, so that $\lambda_n > 0$. The homogeneous equation is
$$a_n'' + \lambda_n a_n = 0.$$
Its characteristic polynomial is
$$r^2 + \lambda_n = 0,$$
so
$$r = \pm i\sqrt{\lambda_n}.$$
Thus, the homogeneous solution is
$$a_n^{(h)}(t) = A_n \cos(\sqrt{\lambda_n} t) + B_n \sin(\sqrt{\lambda_n} t).$$

Non-resonant case

Assume that
$$\omega \neq \sqrt{\lambda_n}.$$
We look for a particular solution of the form
$$a_n^{(p)}(t) = C\cos(\omega t).$$
Then
$$(a_n^{(p)})''(t) = -C\omega^2 \cos(\omega t).$$

Substituting into the equation

$$a_n'' + \lambda_n a_n = c_n \cos(\omega t),$$

gives

$$-C\omega^2 \cos(\omega t) + \lambda_n C \cos(\omega t) = c_n \cos(\omega t).$$

Factoring $\cos(\omega t)$, we obtain

$$(\lambda_n - \omega^2) C \cos(\omega t) = c_n \cos(\omega t),$$

hence

$$C = \frac{c_n}{\lambda_n - \omega^2}.$$

Therefore

$$a_n^{(p)}(t) = \frac{c_n}{\lambda_n - \omega^2} \cos(\omega t),$$

and the general solution is

$$a_n(t) = A_n \cos(\sqrt{\lambda_n} t) + B_n \sin(\sqrt{\lambda_n} t) + \frac{c_n}{\lambda_n - \omega^2} \cos(\omega t).$$

Key Result

If $\omega \neq \sqrt{\lambda_n}$, then

$$a_n(t) = A_n \cos(\sqrt{\lambda_n} t) + B_n \sin(\sqrt{\lambda_n} t) + \frac{c_n}{\lambda_n - \omega^2} \cos(\omega t).$$

Resonant case

Assume now that

$$\omega = \sqrt{\lambda_n}.$$

In this situation, the forcing frequency coincides with the natural frequency of the mode. The previous ansatz no longer works, because $\cos(\omega t)$ already appears in the homogeneous solution. We therefore look for a particular solution of the form

$$a_n^{(p)}(t) = Ct \sin(\omega t).$$

We compute

$$(a_n^{(p)})'(t) = C \sin(\omega t) + C\omega t \cos(\omega t),$$

and

$$(a_n^{(p)})''(t) = 2C\omega \cos(\omega t) - C\omega^2 t \sin(\omega t).$$

Since $\lambda_n = \omega^2$, substituting into the equation gives

$$a_n'' + \lambda_n a_n = \big(2C\omega \cos(\omega t) - C\omega^2 t \sin(\omega t)\big) + \omega^2 Ct \sin(\omega t).$$

The two terms in $t \sin(\omega t)$ cancel, hence

$$a_n'' + \lambda_n a_n = 2C\omega \cos(\omega t).$$

Comparing with

$$c_n \cos(\omega t),$$

we obtain

$$2C\omega = c_n, \qquad C = \frac{c_n}{2\omega}.$$

Thus

$$a_n^{(p)}(t) = \frac{c_n}{2\omega} t \sin(\omega t),$$

and the general solution is

$$a_n(t) = A_n \cos(\omega t) + B_n \sin(\omega t) + \frac{c_n}{2\omega} t \sin(\omega t).$$

Key Result

If $\omega = \sqrt{\lambda_n}$, then

$$a_n(t) = A_n \cos(\omega t) + B_n \sin(\omega t) + \frac{c_n}{2\omega} t \sin(\omega t).$$

The amplitude grows linearly in time.

3.2 Influence of the Forcing Frequency ω

We now See how the response changes with the forcing frequency.

Case 1: ω far from eigenfrequencies

If

$$|\lambda_n - \omega^2| \geq \delta > 0,$$

then

$$\left| \frac{c_n}{\lambda_n - \omega^2} \right| \leq \frac{|c_n|}{\delta},$$

so the amplitude remains bounded.

Key Result

If ω is far from all natural frequencies, the solution remains uniformly bounded in time.

Case 2: near resonance

If

$$\omega^2 \approx \lambda_n,$$

then the denominator $\lambda_n - \omega^2$ becomes very small and

$$\left| \frac{c_n}{\lambda_n - \omega^2} \right|$$

becomes large.

Key Result

If ω is close to $\sqrt{\lambda_n}$, the amplitude becomes large. This is the near-resonance regime.

Case 3: exact resonance

If

$$\omega = \sqrt{\lambda_n},$$

then

$$a_n(t) \sim t \sin(\omega t),$$

so the amplitude grows linearly in time.

Key Result

When $\omega = \sqrt{\lambda_n}$, resonance occurs and the amplitude grows linearly in time.

Case 4: very large frequency

If $\omega \to +\infty$, then

$$\frac{c_n}{\lambda_n - \omega^2} \sim -\frac{c_n}{\omega^2},$$

so the response tends to zero like ω^{-2}.

Key Result

For very large forcing frequency, the modal response behaves like

$$|a_n(t)| \sim \frac{1}{\omega^2},$$

hence the response is very small.

Remark 5. *We expect the strongest resonance to occur for the first positive eigenvalue λ_1, since it corresponds to the lowest nonzero natural frequency*

$$\omega_1 = \sqrt{\lambda_1}.$$

Indeed, in the non-resonant case, the modal amplitude contains the factor

$$\frac{c_n}{\lambda_n - \omega^2}.$$

When ω^2 approaches λ_n, this denominator becomes small, so the response becomes large. Since λ_1 is the smallest positive eigenvalue, it is the first one that can be approached by the forcing frequency among all oscillatory modes.

The eigenvalue $\lambda_0 = 0$ is associated with constant eigenfunctions, which do not represent oscillatory modes of the system. Therefore, it does not contribute to resonance phenomena in the wave equation.

As a result, the dominant resonant behavior is expected near

$$\omega \approx \sqrt{\lambda_1}.$$

3.3 Proof that the Forcing Term Belongs to $L^2(\Omega)$

The forcing term used in the numerical simulations is

$$f(x, y, t) = e^{-16(x^2+y^2)} \cos(\omega t).$$

For each fixed time $t \geq 0$, the factor $\cos(\omega t)$ is a real constant with

$$|\cos(\omega t)| \leq 1.$$

Hence
$$|f(x,y,t)|^2 = e^{-32(x^2+y^2)}\cos^2(\omega t) \le e^{-32(x^2+y^2)} \le 1.$$
Since Ω is bounded, its measure $|\Omega|$ is finite. Therefore
$$\int_\Omega |f(x,y,t)|^2\,dx\,dy \le \int_\Omega 1\,dx\,dy = |\Omega| < +\infty.$$
Thus, for every fixed time $t \ge 0$,
$$f(\cdot,t) \in L^2(\Omega).$$

Key Result

For every fixed time t, the forcing term
$$f(x,y,t) = e^{-16(x^2+y^2)}\cos(\omega t)$$
belongs to $L^2(\Omega)$ because Ω is bounded and f is bounded.

Remark 6. *If needed, one can also note that*
$$f(\cdot,t) \in L^\infty(\Omega),$$
hence automatically
$$f(\cdot,t) \in L^2(\Omega)$$
because Ω is bounded.

3.4 Resonance Analysis in the Wave Equation

The modal amplitude $a_n(t)$ satisfies $a_n'' + \lambda_n a_n = c_n\cos(\omega t)$.

- **Non-Resonance ($\omega \neq \sqrt{\lambda_n}$):** The solution is bounded: $a_n(t) \sim \frac{c_n}{\lambda_n - \omega^2}$.
- **Resonance ($\omega = \sqrt{\lambda_n}$):** The solution grows linearly: $a_n(t) \sim \frac{c_n}{2\omega} t\sin(\omega t)$.
- **High Frequency ($\omega \to \infty$):** The response vanishes: $|a_n(t)| \sim \frac{1}{\omega^2}$.

4 Finite Element Discretization

To approximate the Neumann eigenvalue problem numerically, we consider a continuous finite element space $V_h \subset H^1(\Omega)$ [3].

The discrete weak eigenvalue problem is: find $(\phi_h, \lambda_h) \in V_h \times \mathbb{R}$, $\phi_h \neq 0$, such that
$$a(\phi_h, v_h) = \lambda_h\, m(\phi_h, v_h) \qquad \forall v_h \in V_h.$$
Choosing a basis $(\varphi_i)_{i=1}^N$ of V_h, we write
$$\phi_h = \sum_{j=1}^N \Phi_j \varphi_j.$$
Substituting into the variational formulation and testing with $v_h = \varphi_i$ gives
$$\sum_{j=1}^N \Phi_j \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i\,dx = \lambda_h \sum_{j=1}^N \Phi_j \int_\Omega \varphi_j\varphi_i\,dx.$$

This leads to the generalized matrix eigenvalue problem (see [3], Eq. (5.1.8))

$$K\Phi = \lambda_h M\Phi,$$

where

$$K_{ij} = \int_\Omega \nabla\varphi_j \cdot \nabla\varphi_i \, dx, \qquad M_{ij} = \int_\Omega \varphi_j\varphi_i \, dx.$$

Key Result

The matrices K and M are the discrete version of the bilinear forms $a(\cdot,\cdot)$ and $m(\cdot,\cdot)$, and the problem

$$K\Phi = \lambda_h M\Phi$$

is the finite-dimensional counterpart of the continuous spectral problem.

Thus, the numerical spectral problem is the finite-dimensional analogue of the continuous variational formulation.

For the time-dependent wave equation, the finite element discretization yields the semi-discrete system

$$M\ddot{U}(t) + KU(t) = F(t),$$

where:

- $U(t) \in \mathbb{R}^N$ is the vector of nodal unknowns,
- M is the mass matrix,
- K is the stiffness matrix,
- $F(t)$ is the load vector, defined by

$$F_i(t) = \int_\Omega f(x,t)\varphi_i(x)\, dx.$$

Remark 7. *This system is the discrete analogue of the modal decomposition*

$$a_n''(t) + \lambda_n a_n(t) = c_n(t),$$

where the eigenvalues of $M^{-1}K$ approximate the continuous eigenvalues λ_n of the Neumann Laplacian.

4.1 Time Discretization of the Wave Equation

To solve the semi-discrete wave equation

$$M\ddot{U}(t) + KU(t) = F(t),$$

we use a centered explicit finite difference scheme in time.

Let $t_n = n\Delta t$ and denote by $U^n \approx U(t_n)$.

Using the second-order approximation

$$\ddot{U}(t_n) \approx \frac{U^{n+1} - 2U^n + U^{n-1}}{\Delta t^2},$$

we obtain

$$M\frac{U^{n+1} - 2U^n + U^{n-1}}{\Delta t^2} + KU^n = F^n.$$

Multiplying by Δt^2 and rearranging gives

$$U^{n+1} = 2U^n - U^{n-1} + \Delta t^2 M^{-1}(F^n - KU^n).$$

Key Result

The explicit time-stepping scheme is

$$U^{n+1} = 2U^n - U^{n-1} + \Delta t^2 M^{-1}(F^n - KU^n).$$

Initialization

Given the initial conditions

$$U^0 = 0, \qquad \dot{U}^0 = 0,$$

we compute the first step using

$$U^1 = U^0 + \Delta t \dot{U}^0 + \frac{\Delta t^2}{2} M^{-1}(F^0 - KU^0),$$

which reduces to

$$U^1 = \frac{\Delta t^2}{2} M^{-1} F^0.$$

Stability condition

To analyze stability, we consider the homogeneous system

$$M\ddot{U} + KU = 0.$$

Projecting onto the eigenbasis of $M^{-1}K$, each mode satisfies

$$\ddot{y}_n + \lambda_n y_n = 0.$$

Applying the time discretization gives the recurrence

$$y_{n+1} - (2 - \lambda_n \Delta t^2) y_n + y_{n-1} = 0.$$

Looking for solutions of the form $y_n = r^n$ leads to

$$r^2 - (2 - \lambda_n \Delta t^2) r + 1 = 0.$$

The scheme is stable if $|r| \leq 1$, which holds if

$$\lambda_n \Delta t^2 < 4.$$

Therefore, the stability condition is

$$\Delta t < \frac{2}{\sqrt{\lambda_{\max}}},$$

where $\lambda_{\max}$ is the largest eigenvalue of $M^{-1}K$.

Key Result

The explicit scheme is stable if

$$\Delta t < \frac{2}{\sqrt{\lambda_{\max}(M^{-1}K)}}.$$

Remark 8. *This condition is the discrete counterpart of the modal stability constraint*

$$\omega \Delta t < 2,$$

where $\omega = \sqrt{\lambda_n}$ is the natural frequency of each mode.

Remark 9. *In practice, $\lambda_{\max}(M^{-1}K)$ is estimated numerically (e.g by power iteration), and the time step is chosen slightly below the stability threshold to ensure robustness.*

5 Numerical Results

5.1 Finite element approximation of the Neumann spectrum

We approximate the Neumann eigenvalue problem using a continuous finite element method of order 3. The discrete problem is the generalized eigenvalue problem

$$K\Phi = \lambda_h M\Phi,$$

where M is the mass matrix and K is the stiffness matrix.

The first computed eigenvalues are

$$\begin{aligned}
\lambda_0 &\approx 4.18 \times 10^{-14}, \\
\lambda_1 &\approx 8.65 \times 10^{-2}, \\
\lambda_2 &\approx 2.24556, \\
\lambda_3 &\approx 2.48009, \\
\lambda_4 &\approx 2.48009, \\
\lambda_5 &\approx 2.62181, \\
\lambda_6 &\approx 4.96028, \\
\lambda_7 &\approx 4.96028, \\
\lambda_8 &\approx 5.56778, \\
\lambda_9 &\approx 9.86961, \\
\lambda_{10} &\approx 9.87021, \\
\lambda_{11} &\approx 9.87074.
\end{aligned}$$

As expected for homogeneous Neumann boundary conditions, we obtain

$$\lambda_0 \approx 0,$$

which corresponds to the constant mode.

The first positive eigenvalue λ_1 is much smaller than the next ones. Therefore, the associated frequency

$$\omega_1 = \sqrt{\lambda_1}$$

is expected to dominate the dynamics.

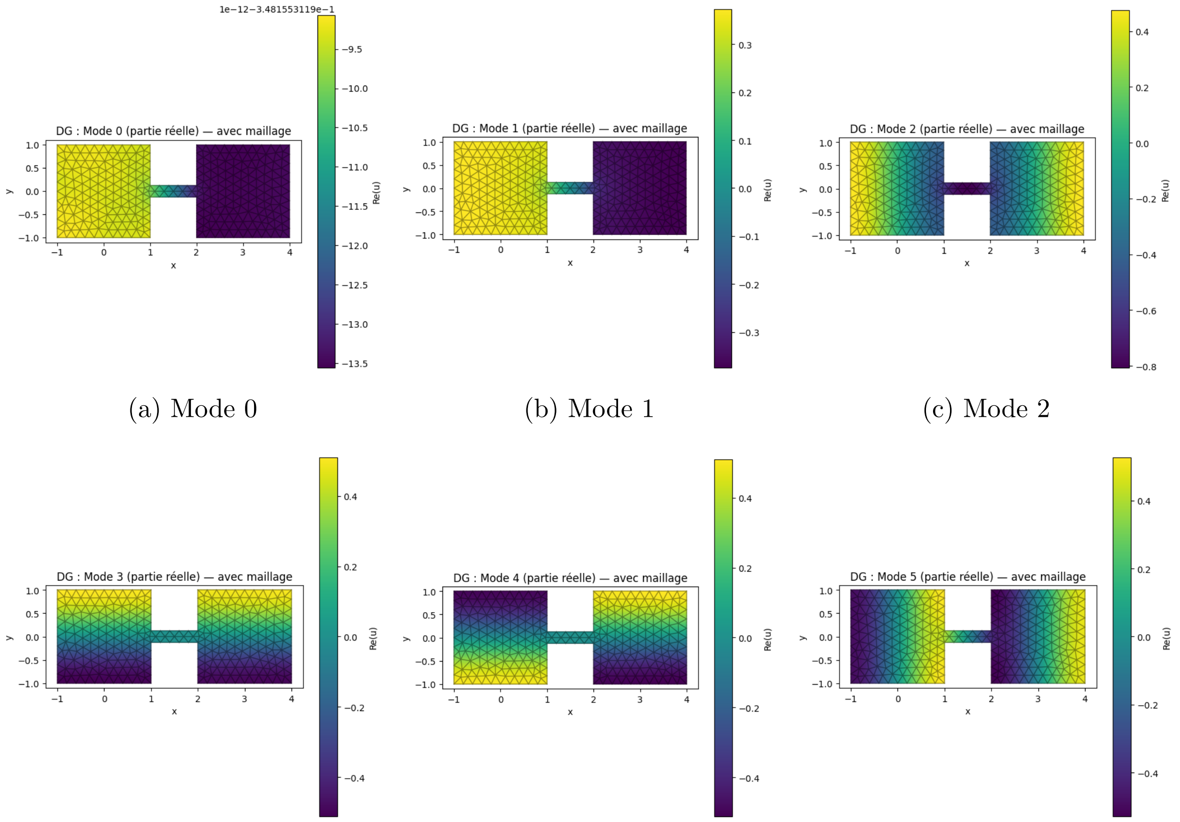
We also observe several clusters of close eigenvalues:

$$\lambda_3 \approx \lambda_4, \qquad \lambda_6 \approx \lambda_7, \qquad \lambda_9 \approx \lambda_{10} \approx \lambda_{11}.$$

This indicates nearly degenerate modes. In particular, since

$$\lambda_3 \approx \lambda_4,$$

we expect the responses associated with ω_3 and ω_4 to be almost identical.



(a) Mode 0 (b) Mode 1 (c) Mode 2

(d) Mode 3 (e) Mode 4 (f) Mode 5

Figure 1: First eigenmodes of the Neumann Laplacian on the dumbbell domain.

5.2 Time discretization of the wave equation

After spatial discretization, the wave equation becomes

$$M\ddot{U}(t) + KU(t) = F(t), \qquad F(t) = b\cos(\omega t),$$

where b is the finite element load vector associated with

$$g(x, y) = e^{-16(x^2+y^2)}.$$

The time integration is performed with the centered explicit scheme

$$U^{n+1} = 2U^n - U^{n-1} + \Delta t^2 M^{-1}(F^n - KU^n).$$

The time step is chosen according to the stability condition

$$\Delta t < \frac{2}{\sqrt{\rho(M^{-1}K)}}.$$

In practice, we use

$$\Delta t = \frac{1.8}{\sqrt{\rho(M^{-1}K)}},$$

which gives

$$\Delta t \approx 0.01493.$$

5.3 Tested frequencies and maximum amplitudes

We test the forcing at the first four nonzero frequencies:

$$\omega_n = \sqrt{\lambda_n}, \qquad n = 1, 2, 3, 4.$$

The corresponding values and the observed maximum amplitudes are given in Table 1.

Mode	Eigenvalue λ_n	Frequency $\omega_n = \sqrt{\lambda_n}$	$\max_{t\in[0,T]} \|U(t)\|_2$
1	$8.6530083870 \times 10^{-2}$	0.29415996	36.1641159
2	2.2455573795	1.49851839	1.64246221
3	2.4800939595	1.57483141	1.64246216
4	2.4800940652	1.57483144	1.64246216

Table 1: Tested frequencies and corresponding maximum amplitudes in the thin corridor case.

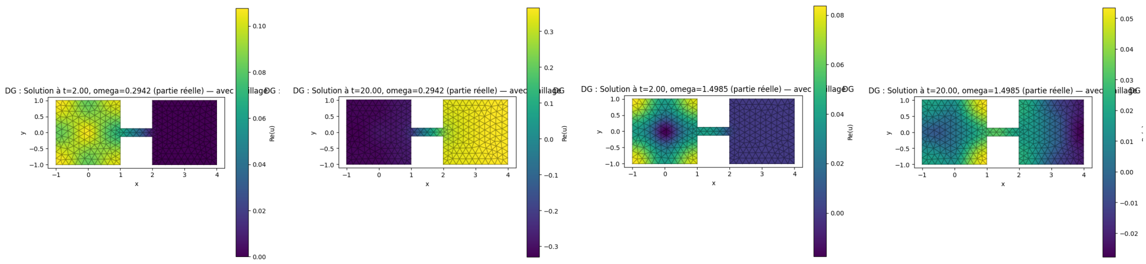
The strongest response is obtained for

$$\omega_1 = \sqrt{\lambda_1}.$$

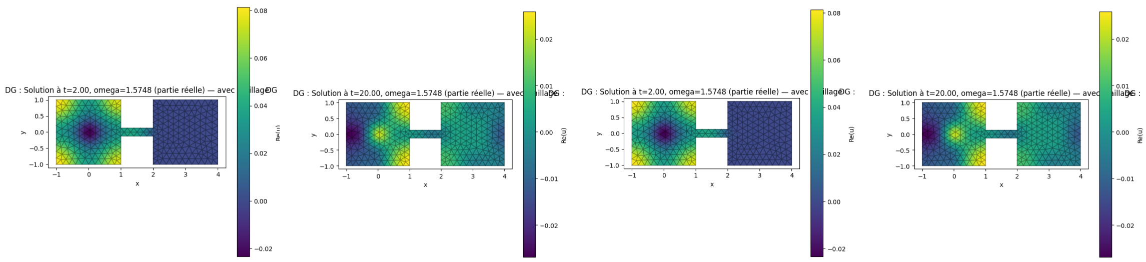
This is exactly what we expected, since λ_1 is the smallest positive eigenvalue. On the contrary, the higher frequencies produce a much weaker response.

5.4 Wave solutions for the tested frequencies

We now show the numerical solution for each tested frequency at an early time and at the final time. This allows us to compare the beginning of the oscillation with the long-time behavior.



(a) $\omega = \sqrt{\lambda_1}$, $t = 2$ (b) $\omega = \sqrt{\lambda_1}$, $t = 20$ (c) $\omega = \sqrt{\lambda_2}$, $t = 2$ (d) $\omega = \sqrt{\lambda_2}$, $t = 20$



(e) $\omega = \sqrt{\lambda_3}$, $t = 2$ (f) $\omega = \sqrt{\lambda_3}$, $t = 20$ (g) $\omega = \sqrt{\lambda_4}$, $t = 2$ (h) $\omega = \sqrt{\lambda_4}$, $t = 20$

Figure 2: Wave solutions for the four tested frequencies in the thin corridor case.

These figures confirm the previous table. The response at

$$\omega_1 = \sqrt{\lambda_1}$$

is much larger than for the other frequencies. The solutions for ω_3 and ω_4 are almost identical, which is consistent with the fact that λ_3 and λ_4 are almost equal.

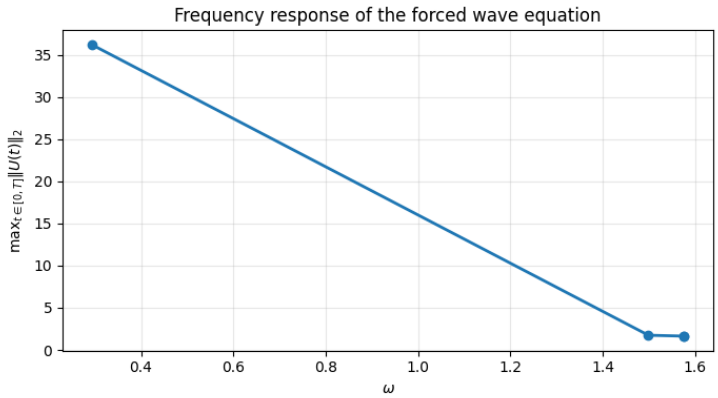


Figure 3: Frequency response of the forced wave equation over the 4 modes.

5.5 Frequency-response curves

For each frequency ω, we define

$$R(\omega) = \max_{t\in[0,T]} \|U(t)\|_2.$$

This quantity is more informative than the final norm, since the solution oscillates and may reach its maximum before the final time, for example here between $t \approx 15$ and $t \approx 17.5$.

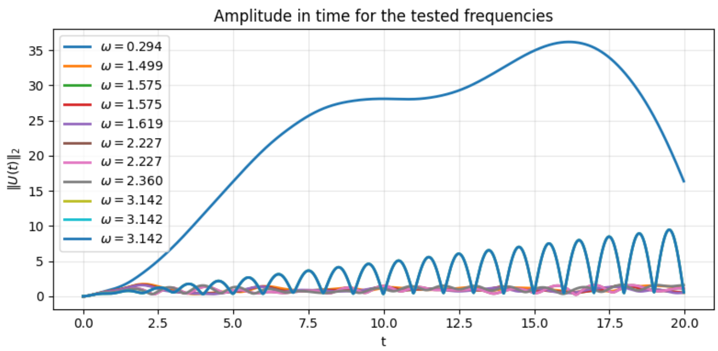


Figure 4: Time evolution of the solution amplitude $\|U(t)\|_2$ for different frequencies ω.

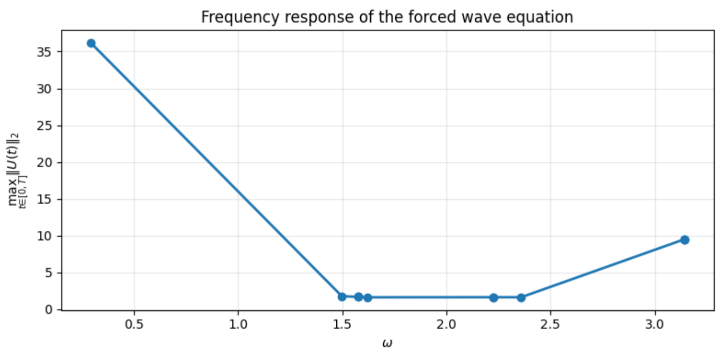


Figure 5: Time evolution of the amplitude $\|U(t)\|_2$ for the tested frequencies.

The left plot shows the evolution of $\|U(t)\|_2$ for each tested frequency. The right plot shows the function

$$\omega \mapsto R(\omega).$$

Both plots clearly indicate that the dominant response is obtained for the first nonzero frequency.

5.6 Projection of the forcing onto the eigenmodes

To understand which modes are excited by the forcing, we project the load vector b onto the eigenmodes:

$$c_k = \langle \phi_k, b \rangle_M = \phi_k^T M b.$$

The quantity $|c_k|$ measures how strongly the forcing excites the k-th mode.

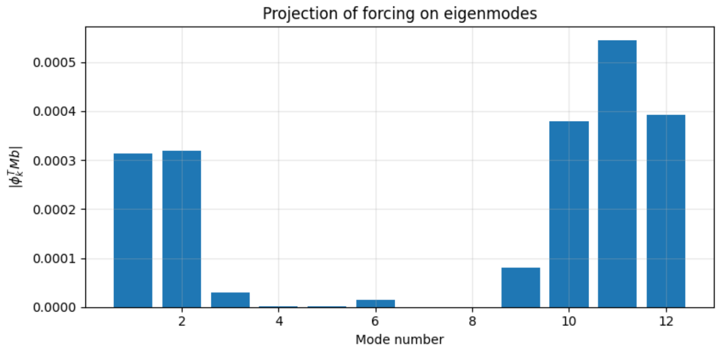


Figure 6: Projection of the forcing onto the first 12 eigenmodes.

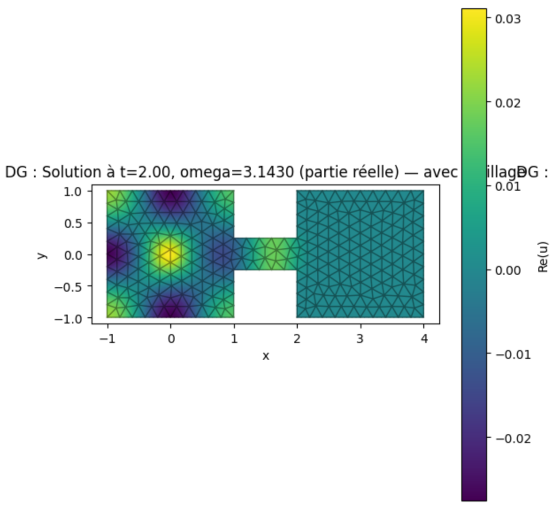
We observe that only a few modes have significant projections. However, a large projection alone does not guarantee a strong response. A mode becomes important only if:

- the projection $|c_k|$ is large,
- and the forcing frequency ω is close to $\sqrt{\lambda_k}$.

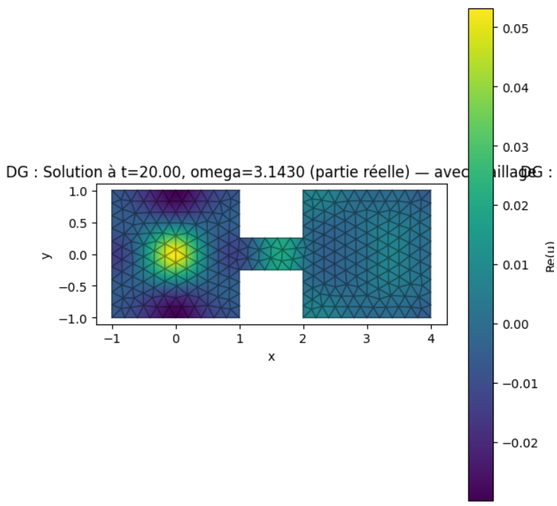
This explains why resonance is strongest near ω_1.

5.7 Additional test at a higher frequency

To illustrate that a higher mode may be weakly relevant even when it is part of the spectrum, we also test the frequency $\omega = \sqrt{\lambda_{10}}$.



(a) Wave solution for $\omega = \sqrt{\lambda_{10}}$ at $t = 2.00$.

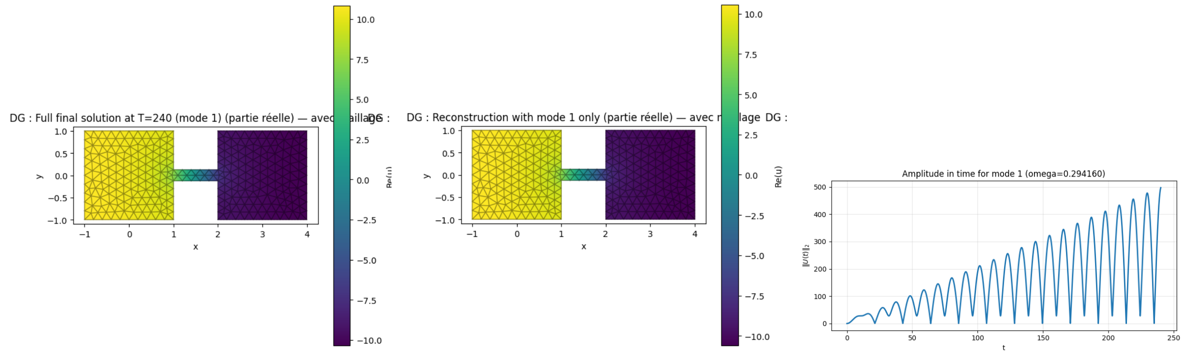


(b) Wave solution for $\omega = \sqrt{\lambda_{10}}$ at $t = 20.00$.

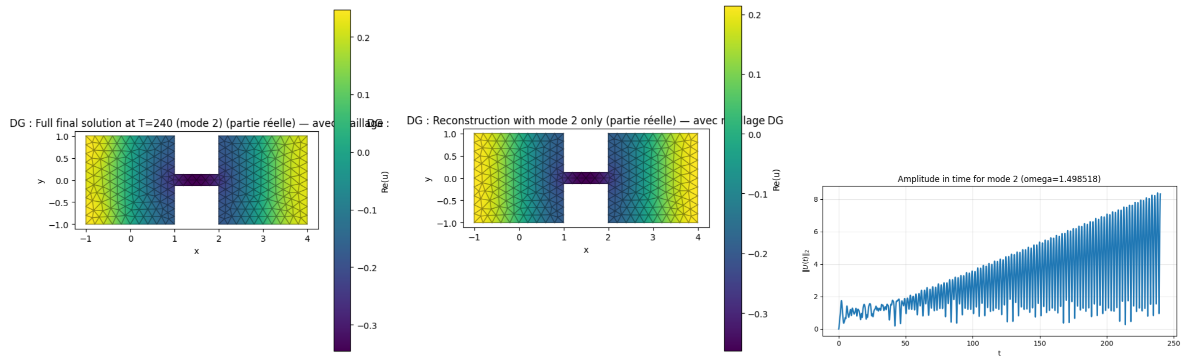
This test confirms that a higher mode does not necessarily dominate the solution, even when its frequency is used. The reason is that the forcing may project weakly onto that mode.

5.8 Modal reconstruction

We now compare the full numerical solution with modal reconstructions at $t = 240$.



(a) Re solution mode1 at $t = 240$. (b) Reconstruction using mode 1. (c) Amp mode1 up to $t = 240$.



(d) Re solution mode2 at $t = 240$. (e) Reconstruction using mode 2. (f) Amp mode2 up to $t = 240$.

Figure 8: Modal reconstruction at $t = 240$ in the thin corridor case.

Mode 1 reconstruction. For mode 1, the final norm is

$$4.974321 \times 10^2,$$

and the relative reconstruction error is

$$2.573667 \times 10^{-2}.$$

This shows that the first mode gives an excellent approximation of the full solution.

For mode 2, the final norm is

$$8.315681,$$

and the relative error with mode 2 only is

$$7.989356 \times 10^{-2}.$$

The reconstruction is still reasonable, but less accurate than for mode 1.

5.9 Thick corridor configuration

We now consider a modified geometry where the corridor height is increased from 0.25 to 0.5.

The first computed eigenvalues are

$$\begin{aligned}
\lambda_0 &\approx 1.07 \times 10^{-12},\\
\lambda_1 &\approx 1.4855 \times 10^{-1},\\
\lambda_2 &\approx 2.0575,\\
\lambda_3 &\approx 2.51398,\\
\lambda_4 &\approx 2.51421,\\
\lambda_5 &\approx 2.75345,\\
\lambda_6 &\approx 5.02982,\\
\lambda_7 &\approx 5.03043,\\
\lambda_8 &\approx 5.20271,\\
\lambda_9 &\approx 9.86961,\\
\lambda_{10} &\approx 9.87826,\\
\lambda_{11} &\approx 9.88419.
\end{aligned}$$

As in the thin case, we again observe close eigenvalues such as

$$\lambda_3 \approx \lambda_4, \qquad \lambda_6 \approx \lambda_7.$$

The stability condition gives

$$\Delta t \approx 0.01713.$$

The tested frequencies and the maximum amplitudes are listed in Table 2.

Mode	Eigenvalue λ_n	Frequency $\omega_n = \sqrt{\lambda_n}$	$\max_{t\in[0,T]} \|U(t)\|_2$
1	$1.4855465967 \times 10^{-1}$	0.38542789	31.8238810
2	2.0574972219	1.43439786	1.81737445
3	2.5139776088	1.58555278	1.80089575
4	2.5142081663	1.58562548	1.80061585

Table 2: Tested frequencies and corresponding maximum amplitudes in the thick corridor case.

The maximum response is still obtained for the first positive frequency, but the value is slightly smaller than in the thin case:

$$31.82 \quad \text{instead of} \quad 36.16.$$

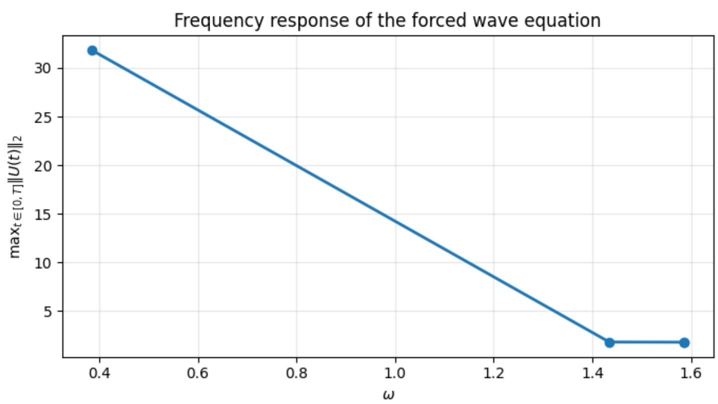
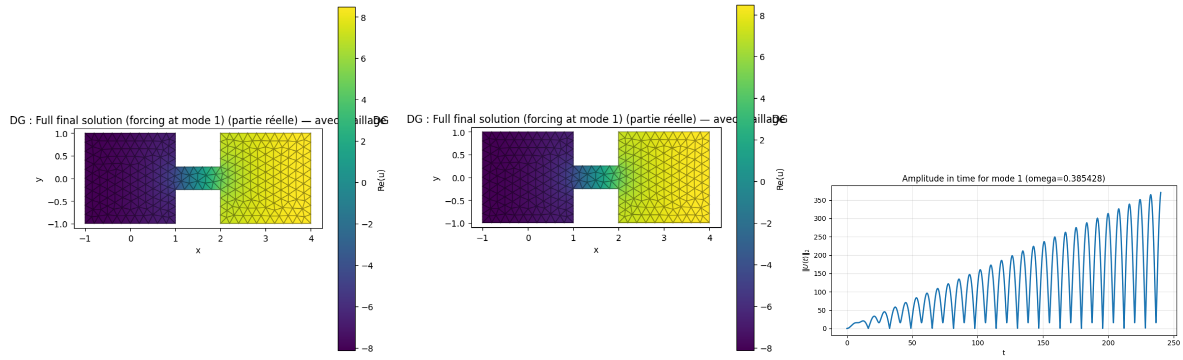
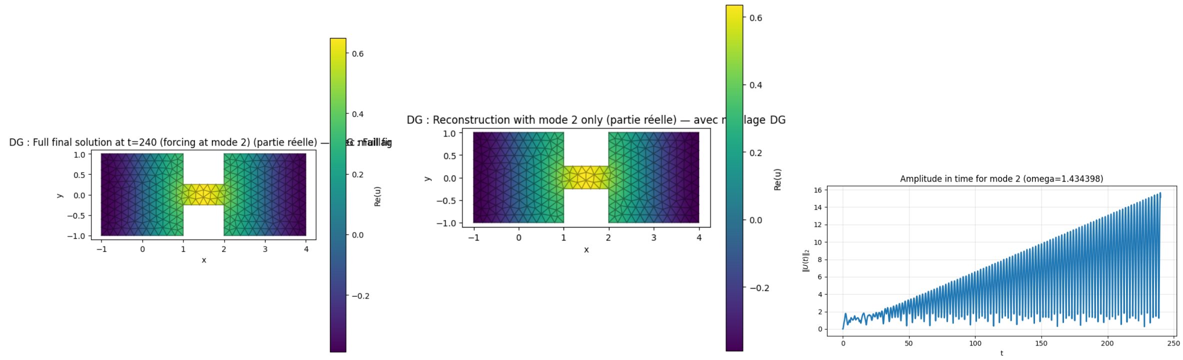


Figure 9: Frequency response of the forced wave equation in the thick corridor case.



(a) Re solution mode1 at $t = 240$. (b) Reconstruction using mode 1. (c) Amp mode1 up to $t = 240$.



(d) Re solution mode2 at $t = 240$. (e) Reconstruction using mode 2. (f) Amp mode2 up to $t = 240$.

Figure 10: Modal reconstruction at $t = 240$ in the thick corridor case.

Mode 1 reconstruction. For mode 1, the final norm is

$$3.707873 \times 10^2,$$

and the relative error is

$$2.501478 \times 10^{-2}.$$

For mode 2, the final norm is

$$1.515058 \times 10^1,$$

and the relative error is

$$3.996329 \times 10^{-2}.$$

5.10 Summary of numerical observations and frequency selection

The numerical results provide a clear picture of the behavior of the dumbbell domain. The spectrum exhibits the expected structure, with $\lambda_0 \approx 0$, a dominant first positive eigenvalue, and set of close eigenvalues.

The time simulations show that the strongest amplification occurs near the first frequency. In the thin corridor case, the maximum amplitude is about 36.16, while in the thick corridor case it decreases to about 31.82. The modal reconstruction confirms that the solution is mainly governed by the first mode, with relative errors around 2.5% in both cases.

Although the solution is computed for twelve excitation frequencies (see Figure 5.5), many of these frequencies produce similar behaviors. For clarity, we focus on a small number of representative frequencies capturing the main regimes: low frequency response, resonance, and high-frequency decay. This is justified since the forcing term mainly excites low frequency modes, while higher modes have a weak contribution. Moreover, the first eigenvalue λ_1 determines the dominant resonance of the system.

Finally, increasing the corridor thickness shifts the spectrum and reduces the resonance effect, highlighting the influence of the geometry on wave propagation, while limiting the number of tested frequencies reduces the computational cost without loss of relevant information.

6 Conclusion

In this project, we studied the link between spectral theory, wave propagation, and finite element approximation on a nontrivial geometry.

The Neumann Laplacian admits a discrete set of eigenvalues and eigenfunctions, leading to a modal decomposition of the wave equation and explaining resonance phenomena. The numerical results confirm that the first nonzero eigenvalue plays a dominant role, with the largest response obtained when the forcing frequency is close to $\omega = \sqrt{\lambda_1}$.

We also observe that close eigenvalues produce similar responses. The comparison between thin and thick corridors shows that the geometry directly influences both the spectrum and the wave behavior, with a reduction of the resonance effect in the thick case.

Finally, the modal reconstruction shows that only a few modes are sufficient to accurately describe the solution.

Overall, this work highlights the effectiveness of spectral analysis for understanding wave phenomena, with applications in areas such as acoustics, vibrations, and fluid flows.

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